

5.4 Green's Theorem_Guide

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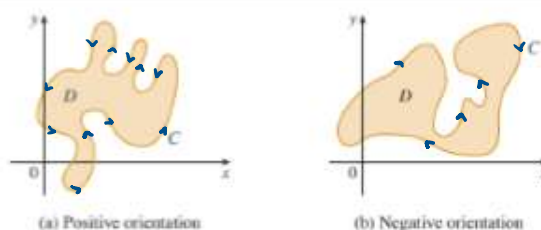


5.4 Green's
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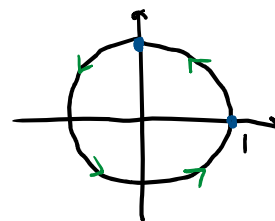
In many ways Green's Theorem is the Climax of MTH 202 because it unites the concepts of partial differentiation, double integrals, and line integrals. In order to fully understand the statement of Green's Theorem we must go over one more concept.

Definition: The positive orientation of a simple closed curve C refers to a single counterclockwise transversal of C . So, if C is given by the vector function $\mathbf{r}(t)$ where $a \leq t \leq b$ and D is the plane region that is bounded by C , then the region D is always on the left as the point $\mathbf{r}(t)$ traverses C .



Example 1: Give a parameterization of the unit circle so that the curve given by your parameterization has positive orientation.

$$\vec{r}(t) = \langle \cos(t), \sin(t) \rangle \text{ where } 0 \leq t \leq 2\pi$$



Theorem (Green's Theorem): Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then
$$\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Notation: The notation $\oint_C P dx + Q dy$ is sometimes used to indicate that the line integral is calculated using the positive orientation of the closed curve C .

Notation: Another notation for the positively oriented boundary curve of D is ∂D . So, one could write:

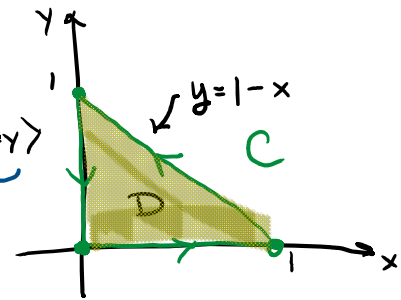
$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_{\partial D} P dx + Q dy.$$

Example 2: Evaluate $\int_C x^4 dx + xy dy$ where C is the triangular curve consisting of the line segments from $(0,0)$ to $(1,0)$, from $(1,0)$ to $(0,1)$, and from $(0,1)$ to $(0,0)$.

We first picture C .

$$\begin{aligned} \text{Note that } \int_C x^4 dx + xy dy &= \int_C \underbrace{\langle x^4, xy \rangle}_{\vec{F}} \cdot \underbrace{\langle dx, dy \rangle}_{d\vec{r}} \\ &= \int_C \vec{F} \cdot d\vec{r} \end{aligned}$$

$$\text{Where } \vec{F} = \langle x^4, xy \rangle$$



Note C is positively oriented, piecewise-smooth, simple, & closed.

P & Q have continuous first ordered partials on $\mathbb{R}^2$ which clearly contains D .

So by Green's Thm

$$\int_C x^4 dx + xy dy \stackrel{?}{=} \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_D (y - 0) dA.$$

$$\text{Note } D = \{(x,y) \mid 0 \leq y \leq 1-x \text{ \& \; } 0 \leq x \leq 1\}$$

$$\text{So } \iint_D y dy dx = \int_0^1 \left[\frac{1}{2} y^2 \right]_0^{1-x} dx = \frac{1}{2} \int_0^1 (1-x)^2 dx = \left[-\frac{1}{6} (1-x)^3 \right]_0^1 = 0 + \left(\frac{1}{6} \right) = \frac{1}{6}$$

While the above shows how Green's Theorem would work, we have yet to provide any sort of justification that this idea is actually valid. Now we provide a partial proof of Green's Theorem.

Theorem (Green's Theorem): Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then $\int_C \underline{P} dx + \underline{Q} dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$.

Green's Theorem is quite difficult to prove in general, but we will give a proof for the special case where the region D is both type 1 and type 2. We will refer to such regions as simple regions.

Partial Proof of Green's Theorem When D is both Type 1 and Type 2:

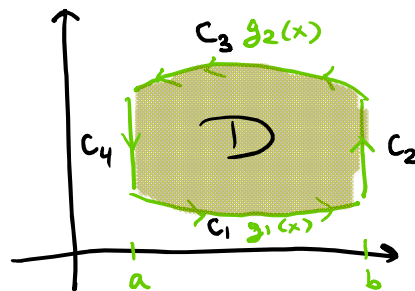
$$\text{Note, if we can show that } \underline{\int_C P dx} = - \iint_D \frac{\partial P}{\partial y} dA \text{ \& \; } \underline{\int_C Q dy} = \iint_D \frac{\partial Q}{\partial x} dA$$

then we will have the desired result.

Since D is a type 1 region we can express it as

$$D = \{(x, y) \mid g_1(x) \leq y \leq g_2(x) \text{ \& } a \leq x \leq b\}$$

Note that C_4 & C_2 may reduce to a point.



$$\begin{aligned} \text{Now we see } \iint_D \frac{\partial P}{\partial y} dA &= \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y} dy dx = \int_a^b P(x, y) \Big|_{g_1(x)=y}^{g_2(x)=y} dx \\ &= \int_a^b [P(x, g_2(x)) - P(x, g_1(x))] dx \end{aligned}$$

$$\text{Now we look at } \int_C P dx = \int_{C_1} P dx + \int_{C_2} P dx + \int_{C_3} P dx + \int_{C_4} P dx$$

In C_2 & C_4 the x -coordinate is constant. So they have parameterizations that look like $\vec{r}_4(t) = \langle a, ? \rangle$ or $\vec{r}_2(t) = \langle b, ? \rangle$

If $x = a$ or $x = b$ then $dx = 0 dt$ & $\int_{C_2} P dx = \int_{C_4} P dx = 0$.

We parameterize C_1 with $\vec{r}_1(t) = \langle t, g_1(t) \rangle$ where $a \leq t \leq b$

We parameterize C_3 with $\vec{r}_3(t) = \langle t, g_2(t) \rangle$ where $a \leq t \leq b$

$$\begin{aligned} \text{So we have: } \int_C P dx &= \int_{C_1} P dx + \int_{C_3} P dx = \int_a^b P(t, g_1(t)) dt + \int_{-C_3} P dx \\ &= \int_a^b P(t, g_1(t)) dt - \int_a^b P(t, g_2(t)) dt \\ &= \int_a^b [P(t, g_1(t)) - P(t, g_2(t))] dt \end{aligned}$$

$$\text{So } \int_C P dx = - \iint_D \frac{\partial P}{\partial y} dA. \quad \text{A similar argument could show that}$$

$$\int_C Q dy = \iint_D \frac{\partial Q}{\partial x} dA. \quad \blacksquare$$

c

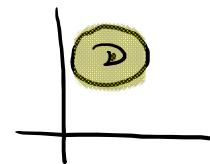
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Example 3: Evaluate $\oint_C (\overset{P}{x^5 + 3y})dx + (\overset{Q}{2x - e^{y^3}})dy$ where C is the circle of radius 2 centered at $(1, 5)$.

Note C is a pos. oriented, piecewise smooth, simple & closed.

P & Q have continuous partials over D .



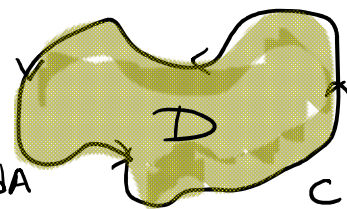
$$\begin{aligned} \text{So by Green's Thm } \oint_C (x^5 + 3y)dx + (2x - e^{y^3})dy \\ &= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_D (2 - 3) dA = \iint_D -1 dA \\ &= - \underbrace{\iint_D 1 dA}_{\text{Area of } D} = -(4\pi) = -4\pi \end{aligned}$$

Example 4: Suppose that C is a positively oriented, piecewise-smooth, simple closed curve in the plane and suppose D is the region bounded by C . TRUE or FALSE: $\frac{1}{2} \oint_C x dy - y dx$ yields the area of D .

$$P = -y \quad \& \quad Q = x$$

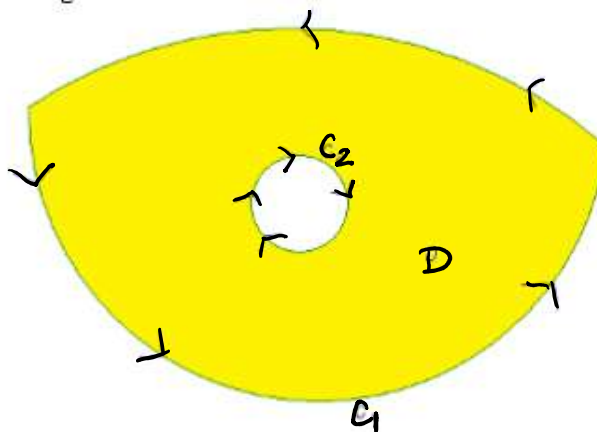
These have cont. first partials.

$$\begin{aligned} \text{By Green's Thm } \frac{1}{2} \oint_C x dy - y dx &= \frac{1}{2} \iint_D (1 - (-1)) dA \\ &= \frac{1}{2} \iint_D 2 dA = \iint_D 1 dA = \text{Area of } D. \end{aligned}$$



A Generalization of Green's Theorem:

Green's Theorem can be generalized to apply to regions with holes, that is, regions which are not simply connected. Consider the following situation:

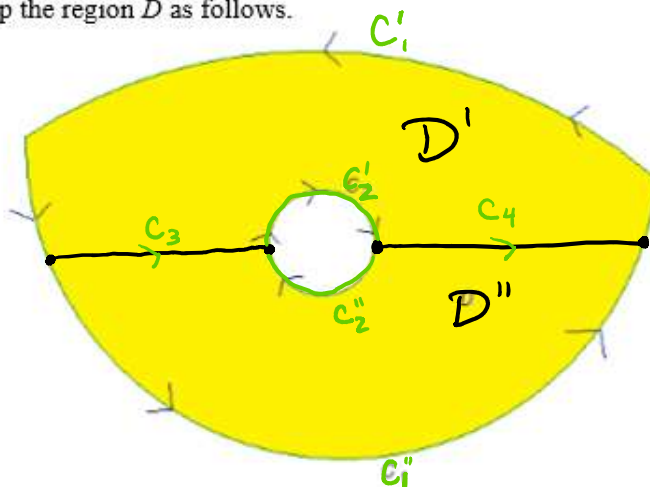


Let C be the curve made up of C_1 & C_2 .

Both these curves are pos. oriented, piecewise-smooth, simple, closed curves.

They bound a region D , that has a hole.

Suppose we want to find $\int_C Pdx + Qdy$ where P and Q have continuous first order partials on a region containing D . To do this we divide up the region D as follows.



Let C' be the curve made from C_1', C_3, C_2' & C_4 .

Let C'' be the curve made from C_1'', C_4, C_2'' & $-C_3$.

C' & C'' both meet the conditions for Green's Thm.

$$\text{So } \int_{C'} Pdx + Qdy = \iint_{D'} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \quad \& \quad \int_{C''} Pdx + Qdy = \iint_{D''} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

We add these equations & get

$$\iint_{D'} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA + \iint_{D''} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \int_{C'} Pdx + Qdy + \int_{C''} Pdx + Qdy$$

$$= \int_{C_1'} + \cancel{\int_{C_3}} + \int_{C_2'} + \cancel{\int_{C_4}} + \int_{C_1''} + \cancel{\int_{-C_4}} + \int_{C_2''} + \cancel{\int_{-C_3}}$$

$$= \int_{C_1'} + \int_{C_2'} + \int_{C_1''} + \int_{C_2''}$$

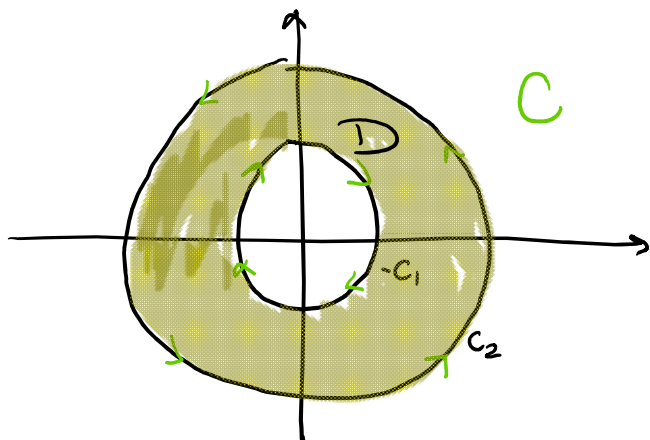
$$= \int_C$$

$$\text{So } \int_C Pdx + Qdy = \iint_{D'} (\quad) dA + \iint_{D''} (\quad) dA = \iint_D (\quad) dA$$

Green's Thm states this!

Example 5: Suppose that C_1 is the unit circle with counterclockwise orientation. Suppose that C_2 is the circle centered at the origin with radius 2 and with counterclockwise orientation. Let C be the positively oriented curve given by $-C_1 \cup C_2$.

a) Sketch a picture of C and the region that it encloses.



b) If $\mathbf{F}(x, y) = \left\langle \overset{P}{\frac{-y}{x^2+y^2}}, \overset{Q}{\frac{x}{x^2+y^2}} \right\rangle$, calculate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

Note C meets the conditions for Green's Thm.

$$\frac{\partial P}{\partial y} = \frac{(x^2+y^2)(-1) - (-y)(2y)}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

$$\frac{\partial Q}{\partial x} = \frac{(x^2+y^2)(1) - (x)(2x)}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

} are cont. over an open region containing D .

By Green's Thm

$$\int_C \vec{F} \cdot d\vec{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_D 0 dA = 0$$